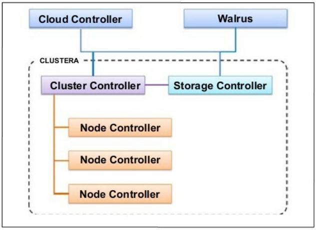

Figure1: Eucalyptus architecture

Resource tagging: This allows users and cloud administrators to assign meaningful metadata to cloud resources. This helps in tracking, as well as eases management and monitoring of specific resource collections used across the cloud.

Maintenance mode: This feature enables cloud administrators to perform maintenance activities on the Eucalyptus cloud without any potential downtime. This helps make sure that the applications hosted on the cloud are always running and that they meet the required SLA levels as well.

Setting up Eucalyptus in your own backyard

In this guide, let's set up a simple Eucalyptus cloud on a set of two machines, one acting as the management server containing the cloud controller, Walrus, the cluster controller and the storage controller, and the other as the node controller running atop a KVM hypervisor.

Figure 2 shows the set-up diagram. In this scenario, we are using two simple desktop machines, each equipped with 4 GB RAM, a 500 GB hard drive and a VT-enabled (Virtualisation Technology) processor on-board. These are the bare minimum requirements to set up a Eucalyptus cloud. However, you can follow the same steps even if you have more machines with better compute capacity between them.

Note: The design can vary as per your requirements, although a minimum of two machines are required to set up the cloud.

Installing the node controller

There are two main ways of going about installing Eucalyptus. The first way is to download the required RPMs onto your machine, install each of them and then manually configure the cloud as per your needs. The second way is much faster and will get your Eucalyptus cloud up and running in a matter of minutes. For this tutorial, let's use the second method, i.e., Eucalyptus Faststart. This is primarily a CentOS-based ISO with all the necessary Eucalyptus

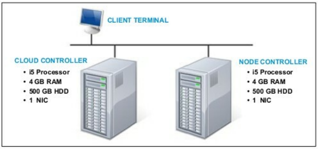

Figure 2: A demo set-up

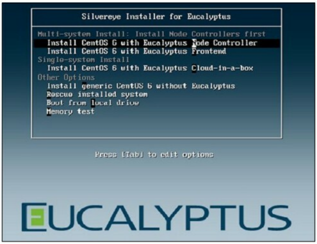

Figure 3: Eucalyptus Faststart boot screen

components embedded in it. All you need to do is burn the ISO to a DVD or USB and run it on your machine. A self-explanatory wizard guides you through the install process and sets up your Eucalyptus cloud for you.

Note: It is recommended that you install the node controllers first so that it becomes easier to add them to the cloud controller once they are all up and ready.

Installing the node controller is a very simple process. Once your machine boots from the Eucalyptus Faststart DVD, select the option 'Install CentOS 6 with Eucalyptus Node Controller' from the boot screen.

Next, select the appropriate 'Language' and 'Keyboard settings' according to your locale.

Provide a 'Static IP' and a suitable 'Host Name' to your node controller in the 'Network Configuration' wizard.

Provide a strong 'Root Password' for your node controller. Once done, your node controller along with the base OS will be installed. You will have to reboot your system once the installation completes.

Log in to the node controller using the root user's username and password. A few scripts run automatically at this time to set the node controller's networking configurations. Once completed, your node controller is ready to be added to a cloud controller.

Note: Follow the above mentioned steps for each of the node controllers that you wish to use for your cloud.